

DATA FLOW & USER STORIES

1. System Actors

Student / User	Register, login, view FAQs/categories, ask questions and receive relevant answers.
Administrator	Manage FAQs, categories and users; update roles and maintain system content.
AI Matching Engine	Process user question text, compare against FAQ information, calculate match score and return relevant results.
MongoDB Database	Store users, categories and FAQ information.

2. Level-0 Data Flow

USER → Web Interface → API Routes → Controllers → AI Matching / Database → Response → Web Interface → USER

3. AI Question Flow

1	User enters a question.	Raw question text
2	AI engine tokenizes and normalizes the text.	Processed tokens
3	Stop words and irrelevant tokens are reduced.	Useful terms
4	Question, keyword and answer matches are calculated.	Match scores
5	Results are sorted by score.	Ranked FAQ matches
6	Configured threshold is checked.	Relevant / no relevant match
7	Best result and suggestions are returned.	Answer + related suggestions

4. User Stories

US-01	Student	register an account	I can access the application.
US-02	Student	login securely	I can use protected features.
US-03	Student	view FAQ categories	I can identify relevant topics.
US-04	Student	view FAQs	I can read available answers.
US-05	Student	ask a question	I can receive a relevant answer.
US-06	Student	see related suggestions	I can explore similar information.
US-07	Admin	create and edit FAQs	I can maintain accurate content.
US-08	Admin	manage categories	I can organize FAQ information.
US-09	Admin	manage users and roles	I can control system access.

US-10	Admin	delete outdated content	I can keep the knowledge base updated.

5. API Requirement Mapping

Authentication	POST /api/auth/register POST /api/auth/login GET /api/auth/me	Account creation and authentication
FAQs	GET/POST /api/faqs GET/PUT/DELETE /api/faqs/:id	FAQ access and management
Categories	GET/POST /api/categories GET/PUT/DELETE /api/categories/:id	Category access and management
Users	GET /api/users GET /api/users/stats DELETE /api/users/:id PUT /api/users/:id/role	Administrative user management
AI	POST /api/ai/ask	Question matching and response